

4(a)	displacement of particles/vibration(s)/oscillation(s) is parallel to/along the direction of energy/propagation	B1
4(b)	period = $1 / 800 (= 1.25 \times 10^{-3} \text{ s})$	C1
	time-base setting = $1.25 \times 10^{-3} / 2.5$	C1
	$= 5.0 \times 10^{-4} \text{ s cm}^{-1}$	A1
4(c)(i)	$I \propto A^2$	C1
	$(I_X / I_Y =) [r_Y / r_X]^2 = [A_X / A_Y]^2$	C1
	ratio $A_Y / A_X = 120 / 30$ $= 4.0$	A1

Question	Answer	Marks
4(c)(ii)	1. $v = f \lambda$	C1
	minimum $\lambda = 330 / (800 + 16) = 0.40 \text{ m}$	A1
	2. $f_o / f_s = v / (v - v_s)$ $816 / 800 = 330 / (330 - v_s)$	C1
	$v_s = 6.5 \text{ m s}^{-1}$	A1

Question	Answer	Marks
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